AI 101 for Mechanical Engineers · Module 9

Trees and ensembles

Lesson 9.2

What you will learn

> Explain how decision trees partition inputs, how random forests and boosting combine models, and why learned boundaries require machine- and regime-specific validation.

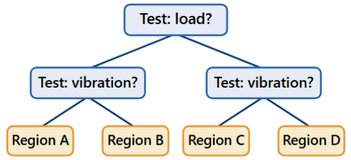
Lesson overview

Module 9 · Lesson 2

Trees and ensembles

AI 101 for Mechanical Engineers

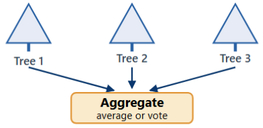
1 A decision tree partitions inputs



Feature tests create **piecewise** prediction regions. Each terminal region gives its own prediction.

A learned threshold is an empirical rule, not a physical fault limit or manufacturer setting.

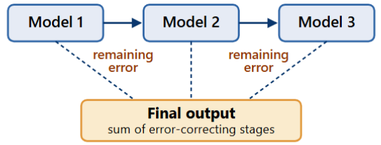
2 A random forest aggregates trees



Randomized trees built **in parallel** (independently, not necessarily at the same time); their predictions are combined.

Can reduce variance, but adds computational cost and is harder to interpret.

3 Boosting corrects errors in sequence



Each stage depends on the errors left by earlier stages: **sequential**, not parallel.

Model complexity must be checked with strict validation.

Connect each method to its practical concern

Method	Concern
Small tree	Easy to inspect, but limited in what it can represent
Deep tree	Can fit training-specific patterns
Random forest	More computation; harder to interpret
Boosting	Complexity needs strict validation

Engineering limits

A rule learned from load and vibration is not proof of a causal physical limit.

Validate across the target operating regimes and equipment. Synthetic examples do not validate industrial hardware.

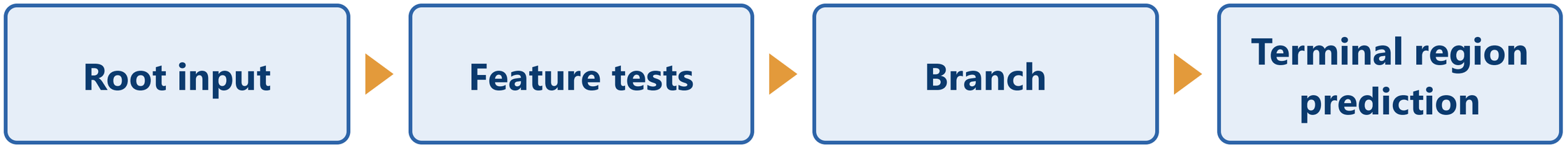
Key takeaway

Trees partition, forests average in parallel and boosting corrects in sequence.

Adding models alone does not establish a better engineering decision.

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A decision tree partitions the input space



- Tests on load or vibration create prediction regions [16, Ch. 8], [47]
- Piecewise: prediction set separately within each region

Small and deep trees raise different concerns

Small tree

- Permits direct engineering inspection
- Restricted representation

Deep tree

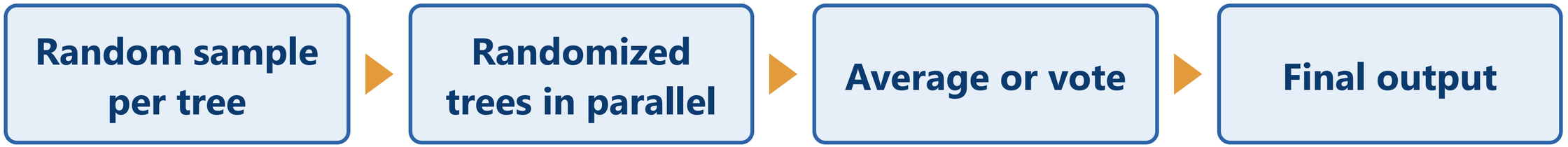
- Uses many partitions
- Can fit training-specific patterns
- Complexity requires validation

> Inspectable tests are not thereby physically valid.

A learned split is not a physical limit

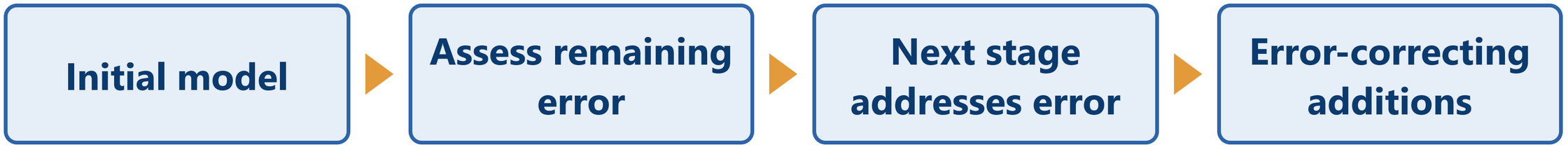
- Tree outputs are piecewise, unlike a smooth response equation
- A learned threshold is an empirical rule from data and training
- Not automatically a physical fault limit or manufacturer specification
- No numerical split threshold is given for the load–vibration example

A random forest aggregates randomized trees



- Combines multiple models into one ensemble [48]
- Can reduce variance; adds cost and can reduce interpretability [49]
- Trees learn from the same data, so predictions are usually correlated [48]

Boosting corrects errors sequentially



- Models are built sequentially, not in parallel [50]
- Each stage depends on earlier stages
- Model complexity must be checked with strict validation

Validate for the target machines and regimes

- A learned load–vibration rule is not proven causal
- Validate across target operating regimes and relevant equipment
- Validation for one regime or unit does not transfer automatically
- Synthetic examples do not validate operational industrial hardware

Each construction has a practical concern

Method	Concern
Small tree	Restricted representation
Deep tree	Training-specific fitting
Random forest	Added cost; harder interpretation
Boosting	Complexity requires validation

> Adding models alone does not establish a better engineering decision.

Availability does not make a method preferable

- RandomForestClassifier and GradientBoostingClassifier are common interfaces [91]
- Neither is automatically preferable
- Compare with a simple baseline; inspect error consequences
- Ask whether results can be validated for the intended machines

CHECK YOUR UNDERSTANDING

Explain one difference between a random forest and boosting.

Answer before moving on.

CHECK YOUR UNDERSTANDING

Explain one difference between a random forest and boosting.

> A random forest aggregates randomized trees constructed in parallel, whereas boosting adds models sequentially to reduce remaining objective errors.

CHECK YOUR UNDERSTANDING

Does a tree split involving vibration establish a manufacturer-approved fault limit?

Answer before moving on.

CHECK YOUR UNDERSTANDING

Does a tree split involving vibration establish a manufacturer-approved fault limit?

> No. It is a learned prediction rule needing validation on relevant machines and regimes. Its boundary is not a manufacturer specification.

Key takeaways

- Trees partition inputs with feature tests; predictions are piecewise
- Forests aggregate parallel trees; boosting adds sequential corrections
- Learned thresholds do not establish causal or manufacturer limits
- Validate on the relevant machines and operating regimes
- Synthetic examples are insufficient for operational hardware

References

[16] G. James, D. Witten, T. Hastie, and R. Tibshirani, An Introduction to Statistical Learning: With Applications in R, 2nd ed. New York, NY, USA: Springer, 2021. doi: 10.1007/978-1-0716-1418-1

[47] L. Breiman, J. H. Friedman, R. A. Olshen, and C. J. Stone, Classification and Regression Trees. Belmont, CA, USA: Wadsworth, 1984.

[48] L. Breiman, “Random forests,” Mach. Learn., vol. 45, no. 1, pp. 5–32, 2001. doi: 10.1023/A:1010933404324

[49] C. Molnar, Interpretable Machine Learning: A Guide for Making Black Box Models Explainable, 3rd ed., 2025. [Online]. Available: https://christophm.github.io/interpretable-ml-book/

[50] J. H. Friedman, “Greedy function approximation: A gradient boosting machine,” Ann. Statist., vol. 29, no. 5, pp. 1189–1232, 2001. doi: 10.1214/aos/1013203451

[91] scikit-learn developers, scikit-learn User Guide and API Reference, ver. 1.9.1. [Online]. Available: https://scikit-learn.org/stable/ Accessed: Oct. 20, 2026.

Numbers match the course reference list. The written lesson is the authoritative text.